Bashir Ahmad *Senior Software Engineer*

bashir.saleh.ahmad12@gmail.com

6710 sands point Dr, Houston, TX 77074

linkedin.com/in/bashir-ahmad-137567372

+1346 469 4135

PROFILE

Results-driven Senior Software Engineer and Azure Solution Architect with 10+ years of experience delivering enterprise-scale, cloud-native platforms across regulated industries. Proven track record in designing secure, compliant systems using Azure (Functions, *API* Management, Logic Apps) and *.NET*-based *microservices*. Experienced with data pipelines, SSO/identity, and integration of procurement tools (e.g., SAP, Coupa, Dow Jones feeds). Skilled in cross-functional stakeholder engagement, *DevOps*, and *agile* delivery in complex, matrixed environments.

SKILLS

Programming Languages: C#, .NET Core, **WPF**, **C/C++**, Java, Python, Node.js, Angular, React.js, Next.js

Front-End Development: React.js, Next.js, Angular, Blazor, Remix, Redux, Zustand, Context | API, Tailwind CSS, Bootstrap, Ant Design, DaisyUI, GSAPReact.js, Next.js, Angular, Blazor, Remix, Redux, Zustand, Context API, Tailwind CSS, Bootstrap, Ant DesigReact.js, Next.js, Angular, | BlazRendering Strategies: SSR, SSG, ISG

Back-End Development: Node.js (Express.js, NestJS), Java (Spring Boot), .NET Core (C#), Python (Django, | Flask)Express.js, NestJS) API Design: RESTful APIs. GraphQL

Database Management: PostgreSQL, MySQL, MongoDB, Microsoft SQL Server, DynamoDB, CosmosDB, | Firebase, Google BigQuery, SupabasePostgreSQL. MySQL. MongoDB, Microsoft SQL ServeORMs: Prisma, | TypcORM, Scquclizc, Hibernate, JPA, SQLAlchemy, Entity Framework, SQL Server, Oracle,

Data Visualization and BI Tools: SAP, Coupa, Dow Jones Risk & Compliance, Procurement Platform Integration | PowerBI, Looker, LookML, Chart.JS, D3.js, AmCharts, Charting Lib, | Draw.io, Miro, UML Diagrams, Virtual Paradigm

Cloud: Azure Functions, *API* Management, Event Grid, Logic Apps, Azure *SQL*, Blob Storage, *Azure DevOps* | Docker, Kubernetes, CI/CD (GitHub Actions), AWS (Lambda, S3, API | Gateway, CloudWatch, SNS, SQS), GCP (Cloud Functions, Tasks), Firebase Docker, Kubernetes, CI/CD (GitHub | Actions), AWS (Lambda, S3, API Gateway, CloudWatch, SNS, SQS), GCP (Cloud Functions, | Tasks), Firebas Docker, Kubernetes, CI/CD (Gitl lub Actions, Azure DevOp | *AWS* (ECS Fargate, *Lambda*, *RDS*, *Cognito*, *API* Gateway, *CloudWatch*) | *Terraform*, *AWS* CodePipeline

Payment Gateway Integration: Stripe, PayPal, Payzone, Plaid

Version Control Systems: Git, Github, Bitbucket, Gitlab

API Interfaces & Docs: GraphQL, RESTful API design, SOAP API, Swagger, JSDoc

Workflows: Pipedream, Hookdeek, Cloudflare, N8N

Authentication: OAuth2, JWT, Role Based Access Control (RBAC), Rate Limiting, Audit Logging SSO, *OAuth2*, *SAML AWS Cognito* | *HIPAA*, *SOC2*, *MuleSoft*

Software Development Practices: Architecture & Methodology: Solution Architecture, Microsoft Cloud Adoption Framework (familiar), GDPR Compliance, TOGAF (working knowledge) | Design and development, Software development lifecycle management, | Research and development, Performance optimization, NX Monorepo, SaaS, Scalable Infrastructure, | Microservices Architecture, System Architecture, Technical Documentation

Project Management: Risk Management Workflows, ESG Integration, GDPR, ISO Standards, Stakeholder Engagement | Tools: JIRA, ClickUp, Trello, Linear | Methodologies: Agile, Scrum, Sprint. Planning. | Kanban | Workflows: Pipedream, N8N, Hookdeck, Cloudflare

UI/UX Design: UI and UX principles

Logging and Observation: Sentry, GlitchTip, Datadog, BetterStack Logtail. Cloudwatch

Testing & Quality Assurance: Jest, Supertest, Playwright, PyTest, JUnit, NUnit, PostmanAutomation & | Standards: ESLint, Prettier. Husky

Quality and Standards: Eslint, Husky, Prettier

Other Skills and Qualities: Multilingual programming, Database Design, Database Migration, Domain | Driven Design (DDD), Mentorship, Design Patterns, Distributed Systems, Project Ownership, Team Lead, | Work Independently, Operational Excellence, Communication Skills, Performance Optimization Techniques

Monitoring & Logging: Datadog, Sentry, Bl.K Stack, Prometheus, GlitchTip, BetterStack Logtail, AWS | CloudWatch

Enterprise Tools & Integration: Enterprise Tools & Integration: SAP, Coupa, Dow Jones Risk & Compliance Feeds, Procurement Platform Integration

PROFESSIONAL EXPERIENCE

Senior Software Engineer Nov 2023 – Present

Texas Label Printers, LLC

- Led full-stack development of data-driven healthcare applications focused on enabling early cancer detection, working at the intersection of clinical workflows, bioinformatics, and real time patient insights. Engineered secure, high-performance systems using Java (Spring Boot). .NET Core, Python (Django/Flask), and Node.js/Express.js, contributing across services that processed and visualized molecular and clinical data for internal scientists and external healthcare partners.
- Developed and maintained microservices and RESTful/GraphQL APIs for clinical trial management, specimen tracking, and genomics data pipelines, utilizing PostgreSQL, MongoDB, Microsoft SQL Server, and Redis for scalable data storage and retrieval.
- Delivered full-stack features using the MERN stack (MongoDB, Express.js, React.js, Node.js) and Next.js, integrating complex datasets into responsive, role based Ul dashboards for clinicians and lab technicians.
- Applied domain-driven design (DDD) and best practices for multi-tenant systems, building extensible modules for RBAC, audit logging, data consent, and HIPAA/GDPR-compliant authentication (OAuth2, JWT).
- Integrated third-party services including Stripe for patient billing, Firebase for push notifications, and internal/external APIs for lab test status and patient data exchange.
- Containerized backend services with Docker and deployed via Kubernetes on AWS/GCP. utilizing Lambda, Cloud Functions, API Gateway, S3, and CloudWatch for monitoring and automation.
- Employed and GitHub Actions to manage CI/CD pipelines, ensuring rapid, reliable deployments across multiple environments (staging, production, clinical sandbox).
- Led initiatives to improve observability and debugging by integrating ELK Stack, Prometheus, and Datadog for logs, metrics, and alerts across distributed systems.
- Built and maintained scalable frontends using React.js, Redux, Tailwind CSS, and Ant Design, supporting dynamic form generation, clinician workflows, and conditional rendering of complex medical data.
- Created unit, integration, and end-to-end test suites with Jest, Supertest, PyTest, and NUnit, and ensured high coverage to meet healthcare grade software standards.
- Acted as a cross-functional engineering lead-collaborating with data scientists, bioinformaticians, and clinicians to define technical priorities aligned with patient outcomes and research goals.
- Architected and deployed *AWS*-native enterprise web portals using ECS (Fargate), *Lambda*, *API* Gateway, and RDS with full integration of *AWS* Cognito for secure user access and Single Sign-On (SSO) via *OAuth2* and SAML protocols.
- Integrated observability tools including *AWS CloudWatch*, *Datadog*, and ELK Stack to enhance system visibility and troubleshoot production issues in real-time.
- Designed infrastructure-as-code (IaC) using Terraform and automated deployments with *AWS* CodePipeline, improving *CI/CD* workflows and operational scalability.
- Coordinated with cross-functional stakeholders to gather architectural requirements and enforce HIPAA compliance across all infrastructure and *APIs*.

Full Stack Engineer Jul 2019 – Sep 2023

Innos

- Designed and implemented an Azure-hosted supplier risk management platform supporting ESG, compliance, and vendor lifecycle workflows.
- Built and deployed *.NET Core microservices* with Azure Functions, *API* Management, Logic Apps, and Azure *SQL* to deliver scalable, compliant architecture.
- Integrated enterprise procurement systems including SAP, Coupa, and Dow Jones Risk & Compliance feeds via custom connectors and secure *API* gateways.
- Collaborated with InfoSec, Data Governance, and compliance teams to ensure solutions adhered to GDPR, *SOC2*, and data residency regulations.
- Led development of data flow diagrams, interface specifications, and operational support documentation.
- Guided internal and offshore teams using *agile* practices, managing *sprint planning*, QA, and vendor coordination to align with business objectives.
- Delivered real-time event-driven data pipelines leveraging Event Grid, Service Bus, and blob storage to ensure timely risk assessments and alerts.
- Contributed to the development of Tempus's Al-driven clinical decision support systems, enabling physicians and researchers to leverage vast clinical and molecular datasets for real-time insights into cancer diagnostics and treatment planning. Designed, implemented, and scaled cloud native full stack applications integrating AI models, patient data workflows, and clinician-facing dashboards.
- Developed secure, scalable microservices using Java (Spring Boot), .NET Core, Python (Django/Flask). and Node.js, powering internal tools and external-facing applications for clinicians, lab techs, and pharmaceutical partners.
- Engineered and optimized GraphQL and REST APIs to serve complex genomic and clinical data to Al inference engines, lab portals, and analytics dashboards-enabling precision treatment decisions at the point of care.
- Built clinician interfaces using React.js, Next.js. Redux Toolkit, and Tailwind CSS, with dynamic visualizations of biomarkers, genomic sequences, and Al-driven treatment recommendations.
- Designed and implemented real-time workflows for diagnostic pipelines using Kafka, RabbitMQ. Redis, and WebSockets, ensuring seamless communication across lab data ingestion and reporting modules.
- Applied domain-driven design and clean architecture principles to backend services supporting multi- tenant logic, role-based access control (RBAC), and audit logging for IIIPAA and GDPR compliance.

- Utilized MongoDB, PostgreSQL. Microsoft SQL Server, and DynamoDB to manage structured and semi- structured clinical/molecular data, applying indexing and caching strategies for performance.
- Lcd integration with internal and external data providers, securely ingesting EMR, lab results, and pathology reports using FHIR-compliant APIs. OpenAPI, and Swagger documentation.
- Containerized services using Docker, and orchestrated deployments with Kubernetes across AWS (Lambda, API Gateway, S3, EC2) and Azure DevOps, enabling scalable, fault-tolerant systems with automated CI/CD pipelines.
- Implemented frontend security and performance enhancements including JWT/OAuth2 authentication, lazy loading, server-side rendering (SSR), and optimized build tooling via Vite, Webpack, and Babel.
- Wrote robust test suites using Jest, Supertest, NUnit, and PyTest, supporting continuous integration with GitHub Actions and Azure DevOps Pipelines for deployment across multiple environments.
- Collaborated with data science and bioinformatics teams to productionize Ml. model outputs, integrating real-Lime predictions into clinician-facing tools using scalable APIs and visualizations.
- Supported clinical operations by building dynamic admin dashboards, patient onboarding portals, and diagnostic reporting interfaces-enabling fast and accurate patient outcomes at scale.
- Delivered secure and scalable healthcare-focused *microservices* with *AWS*-based architecture, including *Lambda* functions, Cognito authentication flows, and ECS-deployed workloads.
- Implemented SSO mechanisms (*OAuth2*/SAML) in collaboration with identity providers, ensuring seamless login experience and secure *API* access for clinicians and pharma partners.
- Designed reusable backend frameworks with middleware integrations (e.g., MuleSoft) for scalable data flow between internal systems and third-party services.
- Focused on healthcare data compliance and governance by applying SOC2 and HIPAA guidelines in cloud-native workflows and access controls.

Software Engineer Feb 2013 – Jul 2019

Devsinc

- Contributed to the development and maintenance of enterprise-grade web applications and internal tools across fintech and healthcare client projects. Collaborated in cross-functional Agile teams to build modular, maintainable software systems while deepening hands-on experience with full-stack development.
- Assisted in building and maintaining backend APIs using Java (Spring Boot) and .NET Core, supporting user authentication, data management, and third-party integrations.
- Participated in the development of responsive user interfaces using React.js. Bootstrap, and Tailwind CSS, improving UX for internal dashboards and client portals.
- Supported integration of MongoDB and PostgreSQL databases using Mongoose, Hibernate, and Entity Framework, contributing to query optimization and schema design.
- Wrote unit and integration tests using JUnit, NUnit, and Postman, improving lest coverage and supporting stable application releases.
- Collaborated in the design and development of RESTful APIs and assisted in initial GraphQL schema design for internal services.
- Deployed containerized applications using Docker and maintained local development environments for streamlined collaboration.
- Gained exposure to CI/CD workflows using GitLab CI and GitHub Actions, contributing to smoother deployment cycles and version control practices.
- Documented technical specifications and participated in code reviews to improve code quality and team learning.
- Worked closely with senior developers to understand system architecture, participate in sprint planning, and troubleshoot production bugs.
- Cloud & Infrastructure: *AWS* (ECS, Fargate, *Lambda*, Cognito, RDS, *API* Gateway, *CloudWatch*, S3, SNS, SQS), Terraform, CodePipeline, CloudFormation
- *Security* & Compliance: *OAuth2*, SAML, SSO, HIPAA, SOC2, Cognito*OAuth2*, SAML, S
- Middleware Integration: MuleSoft, *API* Orchestration
- Observability: *Datadog*, *AWS CloudWatch*, ELK Stack*Datadog*, *AWS CloudWatch*, ELK St
- Architecture & *DevOps*: Enterprise Web Portals, *Microservices*, *Scalable Infrastructure*, Infrastructure as Code (IaC), *CI/CD* Automation (CodePipeline, *GitHub Actions*)

Projects

Integrated Oil & Gas Data Intelligence Platform Nov 2023 – Present

Senior Software Engineer

Technologies

Java (Spring Boot, Hibernate, JPA), Python (Django, Flask, Pandas, NumPy, Scikit-learn), C# (.NET Core, **WPF**, Entity Framework, XAML, MVVM, C++ Interop), MERN Stack (MongoDB, Express.js, React.js, Node.js), SQL Server, PostgreSQL, GraphQL, REST APIs, PowerBI, D3.js, Chart.js, AWS (Lambda, S3, API Gateway), Docker, Kubernetes, Azure DevOps

Led the design and development of a **multi-tier, cross-platform energy analytics platform** built to support the **full oil & gas production lifecycle** — from **geological exploration and seismic data interpretation** to **real-time production monitoring and optimization**. The system integrates **desktop (WPF)**, **web (MERN)**, and **cloud-native** environments into a **unified enterprise-grade solution**, enabling geologists, engineers, and operations teams to access, analyze, and visualize **large-scale geological, reservoir, and production datasets** securely and efficiently. The platform supports **multi-tenant architecture**, **role-based access control**, and **real-time data visualization** for improved decision-making in petroleum engineering workflows.

Key Contributions & Responsibilities:

Java Stack:

- Architected and implemented **Spring Boot microservices** for high-performance ingestion, transformation, and retrieval of geological and wellbore datasets.
- Utilized **Hibernate/JPA ORM** for seamless persistence with **PostgreSQL** and **SQL Server**, optimizing database queries and indexing strategies for large-scale datasets.
- Secured all Java-based services with **JWT/OAuth2 authentication** and role-based access control (RBAC) to comply with industry-grade security standards.

Python Stack:
- Built robust **data processing pipelines** using **Django REST Framework** for API endpoints and **Flask microservices** for lightweight analytical workflows.
- Performed **predictive analytics and reservoir simulation modeling** leveraging **Pandas, NumPy, and Scikit-learn** to forecast production and optimize drilling strategies.
- Integrated Python ML models into backend services for **automated anomaly detection** in production sensor data streams.

C# Stack:
- Designed and delivered **WPF desktop applications** for engineers to **visualize and manipulate geological and production datasets**, following **MVVM architecture** for maintainability and scalability.
- Integrated **C++ modules** into .NET solutions for performance-critical simulation algorithms, enabling faster **reservoir modeling** and production optimization.
- Implemented **real-time 3D geological visualization** with **OpenGL integration** in WPF, allowing engineers to interactively explore subsurface models.

MERN Stack:
- Developed a **React.js web portal** for remote access to geological interpretations, production dashboards, and live well monitoring tools.
- Built backend services using **Express.js** and **Node.js** for **API orchestration, data aggregation, and GraphQL querying**.
- Implemented real-time event-driven updates via **WebSockets** to keep dashboards and maps in sync with live field data.

Cloud & DevOps:
- **Containerized** all services using **Docker** and deployed via **Kubernetes** clusters on **AWS** for high availability and scalability.
- Integrated **AWS Lambda** functions for serverless workflows, **S3** for secure data storage, and **API Gateway** for managed API exposure.
- Implemented **CI/CD pipelines** using GitHub Actions and Azure DevOps to automate testing, security checks, and deployment across multiple environments.

Data Visualization:
- Designed and developed **interactive dashboards** using **PowerBI, D3.js, and Chart.js** for production trend analysis, reservoir performance tracking, and drilling activity visualization.
- Created **geological layer mapping tools** to visually correlate subsurface structures with production data.
- Enabled **predictive drilling recommendations** through advanced visualization of machine learning output.

Impact & Achievements:
- **50% reduction** in geological data processing time via optimized ingestion and transformation pipelines.
- **40% improvement** in engineering workflow efficiency by introducing **intuitive WPF-based desktop tools**.
- Enabled **multi-platform access** (desktop, web, mobile) to real-time oil & gas production insights, improving cross-department collaboration and decision-making.
- Successfully delivered an **enterprise-grade, domain-specific software solution** that integrates **Java, Python, C#, C++, and MERN stacks** in one cohesive ecosystem — setting a new benchmark for data-driven petroleum engineering solutions

EDUCATION

Bachelor's degree, Computer Engineering 2016 | Salt Lake City, Utah
University of Utah